

2 Vector Functions

2.1 Vector Functions

Definition 2.1.1 A vector function is a function which takes in each element in its domain, and returns a vector.

In general, a vector function whose domain is the real numbers, and whose range is the set of 3-dimensional vectors takes the form

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle.$$

Example 2.1.2 If

$$\vec{r}(t) = \langle t^3, \ln(3-t), \sqrt{t} \rangle,$$

find the domain of $\vec{r}$.

Theorem 2.1.3 If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ is a vector valued function, then

$$\lim_{t \rightarrow a} \vec{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle.$$

Example 2.1.4 Find $\lim_{t \rightarrow 0} \vec{r}(t)$ where

$$\vec{r}(t) = \left\langle 1 + t^3, te^{-t}, \frac{\sin t}{t} \right\rangle.$$

Definition 2.1.5 A vector function $\vec{r}(t)$ is said to be continuous at a if

$$\lim_{t \rightarrow a} \vec{r}(t) = \vec{r}(a).$$

Example 2.1.6 Describe the curve defined by the equation

$$\vec{r}(t) = \langle 1 + 2t, 2 - t, -1 + 6t \rangle.$$

Example 2.1.7 Sketch the curve defined by

$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle.$$

Example 2.1.8 Find a vector function that represents the curve of intersection between the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

Example 2.1.9 Find a vector function that represents the curve of intersection of the paraboloid $4y = x^2 + z^2$ and the plane $x = y$.